A. Quantitative Aptitude — Priority 1

1. Percentages

What is it?

Percentage means **“out of 100.”**

If you score 72 out of 100:

Percentage = 72%

Basic formula:

$$Percentage = \frac{Part}{Whole} \times 100$$

Example

A student scores 360 marks out of 500. Find the percentage.

$$\frac{360}{500} \times 100 = 72\%$$

Answer: 72%

Practice

The price of a phone is ₹20,000. It increases by 15%.

What is the new price?

Try: ₹23,000.

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2. Profit & Loss

What is it?

You compare the **Cost Price (CP)** with the **Selling Price (SP)**.

- SP > CP → Profit
- SP < CP → Loss

$$Profit = SP - CP \qquad Profit\% = \frac{Profit}{CP} \times 100$$

Example

A product is bought for ₹800 and sold for ₹1,000.

Profit:

$$1000 - 800 = ₹200$$

Profit percentage:

$$\frac{200}{800} \times 100 = 25\%$$

Answer: 25% profit

Practice

A watch is bought for ₹1,200 and sold for ₹1,500.

Find the **profit percentage**.

Answer: 25%

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3. Ratio & Proportion

What is it?

A ratio compares two quantities.

If:

$A:B=2:3$

then for every 2 units of A, there are 3 units of B.

Example

The ratio of boys to girls in a class is **2:3**.

If there are 20 boys, how many girls are there?

$2 \rightarrow 20$

Therefore:

$3 \times 10 = 30$

Answer: 30 girls

Practice

The ratio of A:B is 4:7.

If A = 28, find B.

Answer: 49

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4. Averages

What is it?

Average is the equal distribution of the total.

$Average = \frac{Sum\ of\ values}{Number\ of\ values}$

Example

Find the average of:

10, 20, 30, 40, 50

Sum:

$10+20+30+40+50=150$

Number of values = 5

$Average = \frac{150}{5} = 30$

Answer: 30

Practice

The average of 5 numbers is 24.

Four of the numbers are:

10, 20, 30, 25

Find the fifth number.

Answer: 35

Because total = 5 × 24 = 120.

Known total = 85.

Missing = 120 − 85 = **35**.

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5. Time & Work

This looks scary initially. It's actually just **work rate**.

If A completes a job in 10 days:

$A's\ 1-day\ work = \frac{1}{10}$

If B completes it in 20 days:

$B's\ 1-day\ work = \frac{1}{20}$

Together:

$\frac{1}{10}+\frac{1}{20}=\frac{3}{20}$

So they finish the job in:

$\frac{20}{3}=6.67\ days$

Example

A can complete a job in 10 days and B can complete it in 15 days.

How long will they take together?

$\frac{1}{10}+\frac{1}{15}$

LCM = 30:

$\frac{3}{30}+\frac{2}{30}=\frac{5}{30}=\frac{1}{6}$

Therefore:

6 days

Practice

A can finish a job in 12 days.

B can finish it in 18 days.

How many days will they take together?

Answer: 7.2 days

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6. Time, Speed & Distance

One of the most important formulas:

$Distance = Speed \times Time$

Therefore:

$Speed = \frac{Distance}{Time} \quad Time = \frac{Distance}{Speed}$

Example

A car travels 240 km in 4 hours.

$Speed = \frac{240}{4} = 60$

Answer: 60 km/h

Practice

A train travels at 80 km/h for 3.5 hours.

How far does it travel?

Answer: 280 km

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7. Simple Interest

Simple Interest is calculated only on the **original principal**.

$SI = \frac{P \times R \times T}{100}$

Where:

- P = Principal
- R = Rate
- T = Time

Example

₹10,000 is invested at 8% simple interest for 2 years.

$SI = \frac{10000 \times 8 \times 2}{100} SI = ₹1600$

Amount:

$10000 + 1600 = ₹11600$

Answer: ₹1,600 interest

Practice

Find the simple interest on ₹5,000 at 10% per year for 3 years.

Answer: ₹1,500

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8. Compound Interest

This is different from simple interest because **interest gets added to the principal**, and future interest is calculated on the increased amount.

$FV = PV(1+r)^n$

$FV = PV(1+r)^n = 1000(1+0.05)20 = \$2,653.30$FV=PV(1+r)n=1000(1+0.05)20=$2,653.30

PVPV

$

PVPV

rr

%

rr

nn

nn

Give feedback

For annual compounding:

$A=P(1+\frac{R}{100})^{T}CI=A-P$

Example

₹10,000 at 10% compound interest for 2 years.

First year:

₹10,000 → ₹11,000

Second year:

₹11,000 → ₹12,100

Therefore:

$CI=12100-10000=₹2100$

Answer: ₹2,100

Practice

₹20,000 is invested at 10% compound interest for 2 years.

Find the compound interest.

Answer: ₹4,200

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9. Number System

This is a **big topic**, not one single formula.

You should understand:

- Natural numbers
- Whole numbers
- Integers
- Prime numbers
- Composite numbers
- Even/odd
- Divisibility
- Remainders
- Factors
- Multiples
- Units digit

Example

Is 7,245 divisible by 5?

A number is divisible by 5 if it ends in:

0 or 5

7,245 ends in 5.

Therefore:

Yes.

Practice

Is **7,392** divisible by 3?

Hint:

Add the digits.

7+3+9+2

Answer: Yes, because the digit sum is 21, which is divisible by 3.

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10. LCM & HCF

HCF

Highest Common Factor

The largest number that divides both numbers.

Example:

Find HCF of 12 and 18.

Factors of 12:

1, 2, 3, 4, 6, 12

Factors of 18:

1, 2, 3, 6, 9, 18

Highest common factor:

6

LCM

Least Common Multiple

The smallest number that is a multiple of both.

Multiples of 4:

4, 8, 12, 16, **20**, ...

Multiples of 5:

5, 10, 15, **20**, ...

So:

LCM = 20

A=4A=4B=6B=6

048120612

LCM=12LCM=12

LCM(4,6)=12LCM(4,6)=12

3 jumps of 4 and 2 jumps of 6 meet at 12

*A*A

*A*A

*B*B

*B*B

Give feedback

Practice

Find the **HCF and LCM** of 18 and 24.

Answer:

HCF = **6**

LCM = **72**

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11. Probability

Probability measures the chance of something happening.

$P(E)=\frac{Favourable\ Outcomes}{Total\ Outcomes}$

Probability ranges from:

$0 \le P(E) \le 1$

Example

A normal dice has 6 sides.

What is the probability of getting a 4?

Favourable outcomes = 1

Total outcomes = 6

$P(4)=\frac{1}{6}$

Answer: 1/6

Practice

A dice is rolled once.

What is the probability of getting an **even number**?

Even numbers:

2, 4, 6

So:

$P=\frac{3}{6}=\frac{1}{2}$

Answer: 1/2

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12. Permutation & Combination

This is where people commonly mix things up.

Permutation = Arrangement

Order matters.

Example:

ABC and BAC are different arrangements.

Formula:

$nPr = n!(n-r)!$

Combination = Selection

Order does NOT matter.

Choosing A and B is the same as choosing B and A.

Formula:

$nCr = n!r!(n-r)!$

Example — Permutation

How many ways can you arrange 3 people A, B and C?

Possible:

ABC

ACB

BAC

BCA

CAB

CBA

Therefore:

6 ways

Or:

$3! = 6$

Example — Combination

You have 5 students and need to select 2.

$5C2 = 5!2!3! = 5 \times 42 = 10$

Answer: 10 selections

Practice

There are 6 people.

How many ways can you **select 3 people**?

Answer:

$6C3=20$